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EXTENDED ABSTRACT

Can AI climate emulators capture statistics of rare weather extremes?

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1. Introduction and objectives

Quantifying the regional statistics of rare weather extremes, such as heatwaves and tropical cyclones, is critical for climate adaptation and mitigation (*1–4*). Computationally expensive, high-resolution physics-based global climate models (GCMs) are central to this task (*5*), but using them to directly sample the rarest events is infeasible because the number of independent samples required scales as $1/p$. Statistical approaches such as extreme value theory (EVT) and resampling methods such as ensemble boosting and rare-event sampling (RES) can extend GCM capabilities, yet each has significant limitations. Recently, attention has shifted to using fast AI climate emulators, such as those described in (*6–10*), to generate large ensembles to quantify extreme-event statistics and internal variability uncertainty for a given GCM and forcing scenario.

For this approach to be valid, emulators must **(i)** remain long-term stable and physically consistent, **(ii)** generate physically plausible "gray swan" events with return periods exceeding the training duration (*11*), and **(iii)** reproduce the correct dynamics and statistics (e.g., return periods) of these gray swan events. Current AI climate emulators of the atmosphere have demonstrated stability and some physical consistency, satisfying **(i)**, but evidence for requirements **(ii)–(iii)** remains mixed. While AI climate emulators have been shown to capture general trends in extreme frequency (*12*), the short duration of the historical record prohibits low-uncertainty analysis of extreme event statistics. We therefore aim to validate the performance of AI climate emulators against a long-duration ground truth data set to determine if they can meet requirements **(ii)** and **(iii)**.

2. Methods and data

Our data sets and methodology are summarized in Fig. 1. We simulate 92,400 years of a stationary climate using PlaSim, an intermediate-complexity GCM, train two stable AI emulators (based on the

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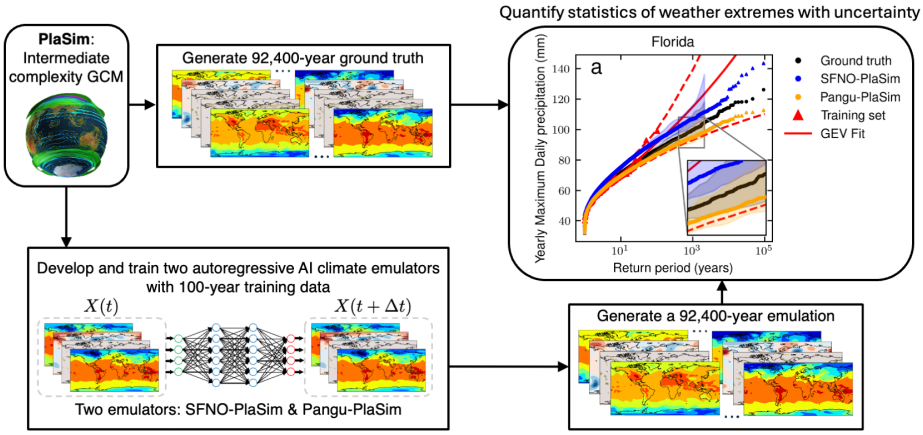

Figure 1. *A schematic of our research methodology described in Sec. 2. The top right panel shows an example return period curve for maximum daily precipitation events over Florida from the unseen ground truth data, training data, and each emulator simulation, as well as a the median GEV fit to the training data and its confidence interval (dashed red line).*

SFNO and PanguWeather architectures) using only 100 years of PlaSim data, and generate 92,400-year emulation ensembles from both models. We then compare the statistics and dynamics of the rarest heatwaves, cold spells, and precipitation events in the ground truth and emulated data sets.

3. Preliminary results and/or expected outcomes

Both SFNO-PlaSim and Pangu-PlaSim are long-term stable ($\geq 2,150$ years), accurately reproduce the climatology and large-scale variability of PlaSim, and can generate extreme events of comparable magnitude to the rarest observed in the long ground truth simulation. However, how well emulators capture the exact event return periods varies by region, event type, and emulator architecture. This is due to several factors, including mean and variability biases in each emulator as well as the lack of land variables, such as soil moisture. Using a simple debiasing based only on the statistics of the 100-year training data set, we find that both emulators more reliably predict 2,000-year return times of extreme weather than GEV fits trained on data of the same duration. This improvement is not universal, however, indicating that further development of frameworks and training techniques for AI climate emulators is needed.

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